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Banking competition in Brazil

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Abstract

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Abstract

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1 Introduction

The Brazilian credit market has undergone significant changes in recent years, driven by new technologies and regulatory reforms that could potentially alter the competitive landscape of the banking sector. This study aims to understand some of the impacts of these changes on the interest rates charged by banks, particularly focusing on the interest rate spread.

Brazilian banking industry is highly concentrated. On December 2024, the five largest banks hold 45.3% total assets of financial institutions, 60.1% of the total loans and 52.8% of total deposits [1]. Even though the concentration is high, it doesn't mean that it is non-competitive, Nakane (2001) tested the level of competitiveness and concluded that Brazilian market doesn't behave as a monopoly/cartel but institutions have some market power.

Santos (2021) used a Cournot model to estimate the effect of different factors on lending rates in Brazil between 2012 and 2016. He concludes that Brazilian rates are higher than other South American countries because of five main factors: IOF tax, high level of risk-free interest rate, higher probability of default, lower recovery rate and more concentrated financial system.

[1] Data from financial institutions accounting reports in https://www.bcb.gov.br/estabilidadefinanceira/balancetesbalancospatrimoniais

2 The model

The model proposed by Ho and Saunders (1981) assumes a market of homogeneous banks working as intermediary dealers. Each bank borrows money from clients to lend it to other clients. Borrowers and lenders arrive randomly on a Poisson distribution, the probability of a new deposit transaction and a new loan transaction are given by:

$$\lambda_D = \alpha - \beta a \tag{2.1}$$

$$\lambda_L = \alpha - \beta b \tag{2.2}$$

Banks can influence the probability of a new deposit or loan by changing it's prices (making it more or less attractive for new clients). The prices for deposits P_D and for loans P_L are:

$$P_D = p + a \tag{2.3}$$

$$P_L = p - b \tag{2.4}$$

where p is the "true" price of the loan or deposit, and a and b are fees that banks can use to increase the probability of a new deposit or loan.

The bank's wealth portfolio (W) is given by $W = Y + I + C$, where Y is the base wealth which is invested in a diversified portfolio, I is the credit inventory and C is the money market position (or short-term net-cash) which is the difference between money market loans borrowings. The expected utility of wealth is given by the following equation:

$$EU(W) = U(W_0) + U'(W_0)r_w W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \tag{2.5}$$

The r_w is the expected rate of return on wealth, σ_I^2 is the variance of the credit inventory, σ_Y^2 is the variance of the base wealth and σ_{IY} is the covariance between I and Y. For each new deposit transaction, the initial credit inventory changes by Q and the credit inventory is $I_0 - Q$. The utility of one deposit transaction is:

$$\begin{aligned} EU(W|\textit{one deposit transaction}) = {} & U(W_0) + U'(W_0)aQ + U'(W_0)r_W W_0 \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 Q^2 + 2\sigma_I^2 QI) \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \end{aligned} \tag{2.6}$$

Similarly, for each new loan transaction, the credit inventory is $I_0 + Q$ and the utility of one loan transaction is:

$$\begin{aligned} EU(W|\textit{one loan transaction}) = {} & U(W_0) + U'(W_0)bQ + U'(W_0)r_W W_0 \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 Q^2 - 2\sigma_I^2 QI) \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \end{aligned} \tag{2.7}$$

$$EU(W|a,b) = \lambda_D EU(W|\textit{one deposit transaction}) + \lambda_L EU(W|\textit{one loan transaction}) \tag{2.8}$$

Deriving the equation 2.8 with respect to the fee a to maximize the wealth, we get to:

$$\frac{\partial EU(W|a,b)}{\partial a} = -\beta\left(U'(W_0)aQ + \frac{1}{2}U''(W_0)\sigma_I^2(Q^2+QI)\right) + (\alpha-\beta a)U'(W_0)Q = 0 \tag{2.9}$$

From 2.9:

$$a = \frac{1}{2}\frac{\alpha}{\beta} + \frac{1}{4}\left(-\frac{U''(W_0)}{U'(W_0)}\right)\sigma_I^2(Q+I) \tag{2.10}$$

Doing the same for the fee b:

$$\frac{\partial EU(W|a,b)}{\partial b} = -\beta\left(U'(W_0)bQ + \frac{1}{2}U''(W_0)\sigma_I^2(Q^2-QI)\right) + (\alpha-\beta b)U'(W_0)Q = 0 \tag{2.11}$$

$$b = \frac{1}{2}\frac{\alpha}{\beta} + \frac{1}{4}\left(-\frac{U''(W_0)}{U'(W_0)}\right)\sigma_I^2(Q-I) \tag{2.12}$$

The credit spread is defined by $S = a + b$. Substituting a and b above for S we get:

$$S = \frac{\alpha}{\beta} + \left(-\frac{U''(W_0)}{U'(W_0)}\right)\sigma_I^2 Q \tag{2.13}$$

Defining the Arrow-Pratt coefficient of absolute risk aversion as $R = -\frac{U''(W_0)}{U'(W_0)}$, we can rewrite the credit spread as:

$$S = \frac{\alpha}{\beta} + \frac{1}{2}R\sigma_I^2 Q \tag{2.14}$$

This is the pure spread that will be estimated in the model, in which the first term α/β is the bank's neutral risk spread, which is the part of the risk that is affected by the competition. The second term is a first-order risk adjustment, which depends on the risk aversion (R), the size of the marginal transactions (Q) and the short-term variance of the interest rate (σ_I^2). The model's assumption of homogeneous banks implies that the terms in the credit spread are the same for all banks, even if the inventories (I) and the base wealth (Y) are different.

The empirical model is built based on the theoretical model above but considering three market imperfections: (i) banks also pay implicit expenses through service fees and other costs indirectly involved in the transactions. (ii) the bank's opportunity cost of holding required reserves and (iii) the default risk on loans, for riskier borrowers, the bank may ask for a higher risk premium on the interest rate. The empirical model estimates the bank margins M given by:

$$M = \delta_0 + \delta_1 IR + \delta_2 OR + \delta_3 DP + u \tag{2.15}$$

The intercept δ_0 is the pure spread defined in equation 2.14, IR is the implicit interest expense, OR is the opportunity cost of the required reserves, DP is the default risk premiums on loans and u is the error term.

The equation 2.15 is the first stage of the model used to find the pure spread δ_0 defined on equation 2.14. In the second stage, the regression will be δ_0 estimated for each time t:

$$\delta_t = \gamma_0 + \gamma_1 \sigma_t^2 + \varepsilon_t \tag{2.16}$$

From equation 2.14:

$$\gamma_0 = \tfrac{\alpha}{\beta} \text{ and } \gamma_1 = \tfrac{1}{2} RQ$$

Competition is affected by the term γ_0 and γ_1 is the risk aversion (which the model assumes as the same for all banks).

3 Clustering Analysis

The econometric model used in this paper assumes homogeneous behavior among financial institutions. However, it is possible that there are different groups of banks with distinct characteristics and behaviors. Farnè and Vouldis (2021) propose a cluster analysis using the k-means method to identify different business models among banks and suggest this as a way to analyse an "increasingly diverse banking industry landscape". Athey and Imbens (2019) more generally suggest "use cluster membership as a way to organize the sample into types of units that may receive different exposures to treatments."

Our objective is to use the clustering analysis to identify groups of financial institutions with similar characteristics and behaviors, and then the econometric model can be applied to each group separately, allowing for a more precise understanding of the dynamics of the banking industry.

The method used is the k-means clustering algorithm, which is a widely used method for partitioning a dataset into K distinct clusters based on the similarity of the data points. Given a set of observations X_i defined by the balance sheet accounts divided by total assets of the i-th financial institution, the goal is partition the feature space into K subspaces (ATHEY; IMBENS, 2019).

3.1 Data preparation

The data used are monthly financial statement accounts of financial institutions obtained from the Banco Central do Brasil (2025). It covers the period from 2010 to 2024 and includes 206 accounts. Most of those accounts are very specific for some institutions, so the data with more than 30% of missing values were excluded, resulting in 56 accounts. The sample was limited to financial institutions with at least 1 billion reais in credit operations. Since the data is monthly, the value of each variable is the average of the 12 months of the year, institutions that didn't have data for all 12 months were excluded of that year. Then, each variable is divided by the total assets of that institution at that year to make the data comparable across institutions of different sizes. Finally, each variable is standarized with mean zero and standard deviation one.

The database has 3210 observations (financial institution per year) and 56 variables. To exclude outliers of each year, we used the Principal Component Analysis (PCA) method. It was calculated the first 4 principal components for each year and the observations with values higher than 4 standard deviations in any of the first 4 principal components were excluded. This resulted in the final database with 3210 observations.

3.2 Principal Component Analysis (PCA)

This method is a multivariate statistical technique for unsupervised learning to reduce the dimensionality of the data while retaining as much information as possible. It does this by finding the directions (principal components) that maximizes the variance explained in the data, and then projecting the data onto these directions. In other perspective, the first principal component is the linear regression that minimizes the orthogonal distance between the data points and the regression line and the other components are defined in the same way but with the restriction that they are orthogonal to the previous components. Ther result is a new set of variables (the principal components) that are uncorrelated and ordered by the amount of variance they explain in the data. The first few principal components can be used to represent the data in a lower-dimensional space, which can be useful for visualization and clustering.

James et al. (2023) define the first principal component as regression line that "minimizes of squared perpendicular distances between each point and the line". It is similar to OLS regression, but instead of minimizing the vertical distances between the data points and the regression line, it minimizes the orthogonal distances. Each subsequent principal component is defined in the same way but with the restriction that it is orthogonal to all previous components. The result is a new set of variables (the principal components) that are uncorrelated and ordered by the amount of variance they explain in the data.

Each principal component has a proprtion of the total variance in the data that it explains, which is called the explained variance ratio. The explained variance ratio is the proportion of the total variance in the data that is explained by each principal component, it can be used to determine how many principal components to retain in order to capture a certain percentage of the variance in the data.

The PCA was used in this study in two different ways: first to exclude outliers and then to reduce the dimensionality of the data before applying the k-means clustering algorithm. To exclude outliers, the database was separated for each year and the variables were normalized to mean zero and standard deviation one. Then the first 4 principal components were calculated for each year and the explained variance ratio of the first four components was very close to 50% on each year. The observations with values higher than 4 standard deviations in any of the first 4 principal components were excluded. This resulted in the final database with 3150 observations.

After excluding outliers, the PCA was used to address the high dimensionality of the data. Steinbach, Ertöz and Kumar (2004) explain that in high dimensionality, the distance between points becomes less meaningful as they become relatively uniform, which can lead to less precise clustering results. In other words, if the financial institutions have many variables (balance sheet accounts), the difference between them become less meaningful and the clusters may not be well defined. The PCA method is an alternative to reduce the dimensionality of the data while retaining as much information as possible.

To reduce the dimensionality of the data, the data was standarized with mean zero and standard deviation one without the year segmentation. The PCA method needs to specify the number of principal components (PCs) to retain, which can be determined by looking at the explained variance ratio. The k-means algorithm also needs to specify the number of clusters K, one way to determine the optimal number of clusters is by calculating the silhouette score, which measures how well each data point fits within its assigned cluster compared to other clusters. The silhouette score ranges from -1 to 1, where a score close to 1 indicates that the data point is well matched to its own cluster and poorly matched to neighboring clusters, while a score close to -1 indicates that the data point is poorly matched to its own cluster and may be better assigned to a neighboring cluster.

The number of principal components retained may affect the clusteing results, so the strategy used here is to find the for each K number of clusters the number of principal components that maximizes the silhouette score. The optimal number of clusters is then determined by looking at the silhouette scores for different values of K and selecting the one that yields the highest score. The results are shown in Figure 1. The chosen K was 5 and the number of principal components retained was 14, which explains 80% of the variance in the data. The silhouette score for this configuration was 0.19, which indicates that the clusters are reasonably well defined, but there is still some overlap between them. The data has 56 variables that the PCA reduced to 14 principal components.

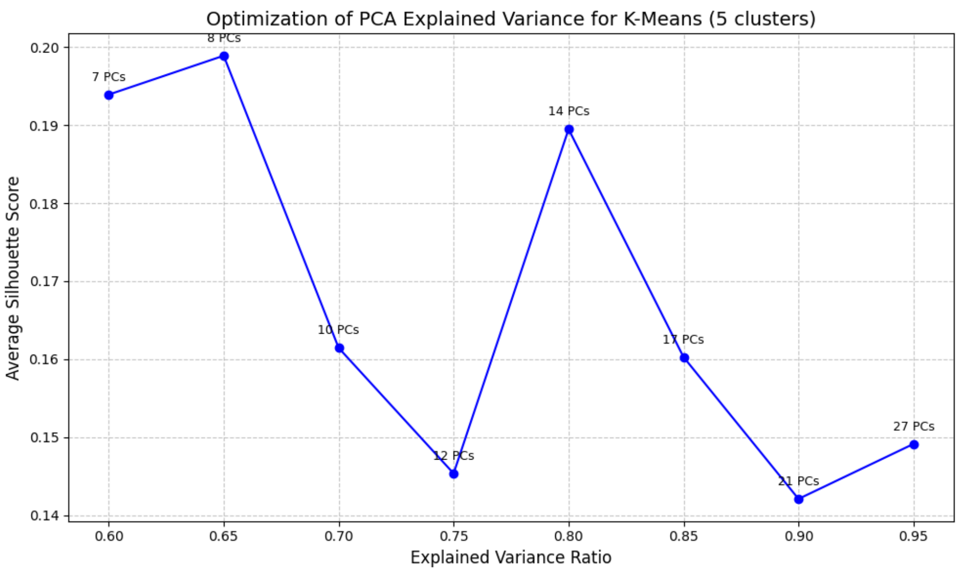

Figure 1 – Optimization of PCA Explained Variance for K-Means (5 clusters)

3.3 k-means clustering

The k-means algorithm aims to partition the data into K clusters by minimizing the within-cluster sum of squares (WCSS), which is defined as the sum of the squared distances between each data point and the centroid of its assigned cluster. The algorithm starts with an initial set of K centroids, which can be randomly selected from the data points or initialized using a specific method. Then, it iteratively assigns each data point to the nearest centroid and updates the centroids by calculating the mean of the data points in each cluster. This process continues until convergence, which occurs when the assignments no longer change or when a maximum number of iterations is reached.

After the parameter definitions, the k-means algorithm is applied to the data projected onto the retained principal components resulting in the 5 clusters. The evolution of the number of institutions in each cluster over time is shown in Figure 2.

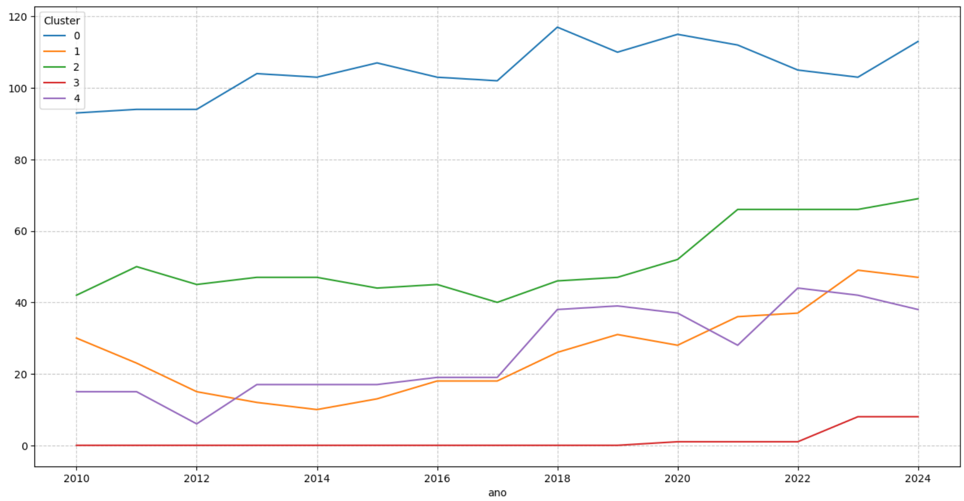

Figure 2 – Evolution of Clusters over Time

Before 2018, one category of financial institutions known as "sociedade de crédito ao microempreendedor e à Empresa de Pequeno Porte" (SCMEPP) did not have it's financial statements disclosed, this changed after a Central Bank's resolution that made the disclosure mandatory (Banco Central do Brasil, 2017). So in the the data we can see a signifiant increase in the number of institutions in cluster 4 after 2018, which is the cluster that contains most of the SCMEPPs. Also in 2018, the, it was created the "sociedade de crédito direto" (SCD), which is a new type of financial institution that operates exclusively through digital platforms and has a more flexible regulatory framework (Banco Central do Brasil, 2018). The SCDs start appearing in the data in 2019 and are only 52 institutions, but they are growing rapidly wth only one in 2019, 34 in 2023 and 39 in 2024.

The cluster 0 is the most stable over time, with a consistent number of institutions throughout the years. Banks are 84.8% of the institutions in this cluster, and they are the largest financial institutions in terms of total assets. Cluster 3 was empty until 2020 and is somposed by SCDs, it represents a very small group of institutions and may be a result of changes in regulation and the emergence of new types of financial institutions.

Clusters 1 and 2

4 Sample and Data

The empirical model is based on Maudos and Guevara (2004). It is a single stage model and the variables used are shown in Table 1.

Dep. Variable:	nim	**R-squared:**	0.4615
Estimator:	PanelOLS	**R-squared (Between):**	0.6613
No. Observations:	3150	**R-squared (Within):**	0.3206
Date:	Tue, May 19 2026	**R-squared (Overall):**	0.4663
Time:	16:26:38	**Log-likelihood**	-1.221e+04
Cov. Estimator:	Clustered		
		F-statistic:	223.03
Entities:	363	**P-value**	0.0000
Avg Obs:	8.6777	**Distribution:**	F(12,3123)
Min Obs:	1.0000		
Max Obs:	15.000	**F-statistic (robust):**	40.263
		P-value	0.0000
Time periods:	15	**Distribution:**	F(12,3123)
Avg Obs:	210.00		
Min Obs:	160.00		
Max Obs:	275.00		

	Parameter	**Std. Err.**	**T-stat**	**P-value**	**Lower CI**	**Upper CI**
const	-26.595	11.046	-2.4077	0.0161	-48.253	-4.9374
crerisk	-0.0145	0.0332	-0.4351	0.6635	-0.0796	0.0507
sd_3m*crerisk	1.0937	0.8717	1.2546	0.2097	-0.6156	2.8029
sd_3anos*crerisk	-0.0122	0.0172	-0.7082	0.4789	-0.0458	0.0215
size	-1.6780	0.6396	-2.6234	0.0087	-2.9321	-0.4238
riskaver	-0.0902	0.0342	-2.6354	0.0084	-0.1573	-0.0231
aoc	2.2743	0.5140	4.4247	0.0000	1.2665	3.2822
ef	-0.0466	0.0146	-3.1855	0.0015	-0.0753	-0.0179
iip	1.3048	0.5021	2.5986	0.0094	0.3203	2.2894
Cluster_0	50.714	11.984	4.2319	0.0000	27.217	74.211
Cluster_1	43.440	11.434	3.7990	0.0001	21.020	65.860
Cluster_2	46.558	12.797	3.6383	0.0003	21.467	71.648
Cluster_4	45.073	10.981	4.1045	0.0000	23.541	66.604

F-test for Poolability: 0.5890
P-value: 0.8758
Distribution: F(14,3123)

Included effects: Time

Variable	Name	Financial Statement Accounts
nim	Net Income Margin	$\frac{Operating\ Revenue - Capital\ Raising\ Cost}{Realizable\ Assets}$
hhi	HHI Index	$\sum_{i=1}^{n} (marketshare_i)^2$
aoc	Average Operating Costs	$\frac{Operating\ Costs}{Total\ Assets}$
riskaver	Risk Aversion	$\frac{\text{Total Equity}}{Total\ Assets}$
sd	Volatility of market interest rates	
crerisk	Credit Risk	$\frac{provisions\ for\ credit\ operations}{Total\ Credit\ Operations}$
*sd*crerisk*	Interaction between credit risk and market risk	$sd \times crerisk$
size	Average size of operations	$\log_{10}(Total\ Credit\ Operations)$
iip	Implicit interest payments	$\frac{Operating\ Costs - Services\ Income}{Total\ Assets}$
reser	Opportunity costs of bank reserves	$\frac{Cash\ and\ Cash\ Equivalents - Interbank\ Liquid\ Investments}{Total\ Assets}$
ef	Implicit interest payments	$\frac{Operating\ Costs}{Operating\ Revenue}$

Table 1 – List of variables used in the empirical model

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Appendix

APPENDIX A – Expected utility of wealth

The expected utility of wealth is given by the equation:

$$E[U(W)] = U(W_0) + U'(W_0)r_w W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \tag{A.1}$$

This function based on the assumption that the agents are risk averse, which is mathematically represented as a concave function, which is defined by the condition: $E[U(W)] < U(E[W])$. Ingersoll (1987) shows that

APPENDIX B – Interest rate volatility estimation with IDkA

$IDkA$ (ANBIMA,) is a set of indices that measure the behavior of synthetic portfolios of federal government bonds with a constant maturity. They are calculated by ANBIMA (Brazilian Association of Financial and Capital Markets Entities) using selected vertices of the yield curve of Brazilian government bonds. The indices are calculated for maturities of 3 months, 1 year, 2 years, 3 years, and 5 years.

The index is calculated using the following formula:

$$IDkA_t = IDkA_{t-1} \frac{(r_{t-1}^{n} + 1)^{\frac{n}{252}}}{(r_t^{n-1} + 1)^{\frac{n-1}{252}}} \tag{B.1}$$

where $IDkA_t$ is the index at time t, $IDkA_{t-1}$ is the index at time $t-1$ and r_t^n is the interest rate of the bond with maturity n at time t. Since the government bonds yield curve past data was not available but $IDkA$ database is disclosed by ANBIMA (ANBIMA,). We used the index index data ($IDkA_t$ and $IDkA_{t-1}$) and the yield curve values (r_t)at 11/04/2025 and manipulated the formula to obtain the interest rate from the previous day:

$$r_t^{n-1} = \left(\frac{IDkA_{t-1}}{IDkA_t} (r_{t-1}^{n} + 1)^{\frac{n}{252}} \right)^{\frac{252}{n}} - 1 \tag{B.2}$$

The, we repeated the process for all the available data to obtain the interest rate time series and then calculated the volatility using the standard deviation of the log returns of the interest rates: